OS Forensics

Operating System Forensics

- **analyzing data stored or processed by an operating system**
  - Involves file systems
    - allocated and unallocated space for deleted or hidden files
    - file/directory structures and registries (and metadata)
  - Event log analysis (configuration/Registry and system-related files)
    - user activity, software configurations, and connected devices
    - login attempts, application activity, and network connections
    - e.g. Prefetch(what/when executed) and LNK Files (what accessed) (Windows)
    - e.g. `~/.bash_history` / , `~/.zsh_history, /var/log/auth.log, /proc/`
  - Memory Dumps
    - volatile data, including active processes and encryption keys
- **with various possible tools**
  - *Exiftool, OSForensics, Autopsy, Volatility* for file recovery,registry analysis, disk signature comparisons, memory analysis

For a rootkit forensics investigation...

- **log analysis**
  - **look for centralized log places**
    - *`/var/log/syslog`, `/var/log/auth.log`*
    - *Windows Event Viewer*
  - rootkits can modify or suppress logs
    - ...but missing logs can indicate their presence
- **kernel activities (since rootkits often operate at kernel level)**
  - analyse kernel modules and system call activities
    - *dmesg*
    - *sysdig, strace*
  - analyse process details
    - *ps, top, procfs*

For a rootkit forensics investigation...

- **memory analysis**
  - memory dump
    - *LIME (Linux Memory Extractor)*
  - memory analysis
    - *Volatility, Rekall*
- **network analysis for hidden or unauthorized network sockets**
  - network traffic
    - *wireshark, netstat, tcpdump, zeek*
  - socket analysis
    - *netstat, ss (socket statistics), lsof*

Memory Forensics

- **examines the contents of volatile memory (RAM) to extract running processes, decrypted data, and user activities. Real-time extraction is complicated by the risk of tampering or data corruption**
- **key points**
  - Volatility of Data
    - RAM holds temporary data that may disappear upon shutdown, making timely capture crucial
  - Memory Dump
    - capturing the state of RAM using tools to analyze it post-mortem
- **tools**
  - Lime, Volatility/Rekall (live acquisition / memory dumps analysis)

Memory "dump"

- a **snapshot of the contents of a computer's Random Access Memory (RAM)**
  - at a specific point in time
- **harvesting**
  - whispers of processes
  - fragments of data,
  - and the faint echoes of applications in motion

Analysis of memory dump

- **recover volatile data**
  - transient data not stored on disk, such as encryption keys and in-memory malware
- **system state understanding**
  - active processes, open files, and network connections
- **malicious activity**
  - analysis can reveal the presence and behavior of malware that operates primarily in memory
- ...

Analysis of memory dump (II)

- **process analysis**
  - Identify suspicious processes like running processes, hidden processes, and anomalies like injected code
  - e.g. a malicious process masquerading as a legitimate system service might be identified through memory analysis
- **code injection detection**
  - locate and analyze injected code or malicious payloads
    - e.g. using plugins such as `malfind` (Volatility) to identify unusual memory regions tied to legitimate processes
  - e.g. detecting a DLL injection in `svchost.exe`.

Analysis of memory dump (III)

- **registry and file system insight**
  - examine registry keys and open files stored in memory
    - malware may affect registry keys for persistence, which may be visible in the memory dump
- **network forensics**
  - analyze active network connections to identify suspicious one, such as exfiltration attempts/connections to malicious servers
- **credential recovery**
  - recover encryption keys and passwords from RAM, decrypting files or communication to access critical evidence
- **crash and failure analysis**
  - understand system crashes or kernel panics, and diagnose vulnerabilities or exploits

Challenges of memory dump (IV)

- **data volume**
  - Memory dumps can be very large, making analysis resource-intensive
- **obfuscation**
  - advanced malware may use techniques like encryption or anti-debugging to evade detection
- **forensic soundness**
  - ensuring that the dump is collected and handled in a way that maintains its integrity is critical for its admissibility in court.

Memory access tools

- **debugging tools**
  - gdb (GNU Debugger) can capture memory regions
  - limited in scope to specific processes
- **crash dump utilities**
  - designed for system administrators to capture memory state
  - debugging purpose but suitable for forensics use
  - e.g. *kdump* (crash dump creation if the kernel crashes)

Memory access tools

- **LiME**
- **Fmem**
  - kernel module to provide a device file for raw physical memory access (`/dev/fmem`)
  - possible direct reading using dd-like tools
- **AVML (Acquire Volatile Memory for Linux)**
  - save recorded images to external locations via Azure Blob Store or HTTP PUT
  - LiME output format (or compressed)

LiME (Linux Memory Extractor)

- **open-source forensic tool designed for live memory acquisition on Linux-based systems**
  - acquisition of the entire contents of a system’s RAM in real time
  - recovering volatile data like passwords, encryption keys, and evidence of running processes
- **a kernel module**
  - interaction at a low system level
    - direct access to memory regions
- **supports remote memory acquisition**
  - extract memory data over a network

LiME

- **LiME maps the entire physical memory of the system, including user space, kernel space, and other volatile data**
- **reads memory contents sequentially and writes them to a specified output destination**
  - file or network
- **Lime format do not alter memory content**
  - no compression, modification, additional metadata
  - consider "endianess" of the architecture
- **output is either stored locally or streamed to a forensic workstation for further analysis**

Volatility

- **open-source memory forensics framework used to analyze memory dumps from various operating systems**
- **designed for digital forensics investigations**
  - uncover evidence
  - identify malicious activity
  - debug systems by examining the contents of a system's memory (RAM)
- **multi-platform**
  - can analise memory dumps from Windows, Linux, macOS, and Android systems
  - from raw dumps, crash dumps, hibernation files, Lime

Volatility technical requirements

- **needs to understand the exact kernel structures of the source OS to correctly parse memory.**
  - without the right profile, every offset calculation is wrong
    - `volatility -f memory.raw imageinfo` can suggest profile information
    - Next command: `volatility -f memory.raw --profile=Win10x64_19041 [pluginname]`
  - Vol3 uses **symbol tables** -
    - JSON files mapping kernel struct offsets for a specific kernel build.
    - Windows, auto-downloads the correct PDB symbols from Microsoft's symbol server.
    - Linux/macOS, you build the ISF (Intermediate Symbol Format) file from the target kernel's debug symbols
    - `python3 vol.py -f memory.raw windows.info`

Volatility plug-in

- **available a rich set of plugins**
- **extract specific information**
  - running processes
  - open network connections
  - loaded kernel modules
  - artifacts like command histories or encryption keys
  - …
- **maintaining forensic soundness**
  - non-destructive analysis (i.e. do not alter original memory dump)

Volatility use cases

- **incident response**
  - investigate malware infections by extracting indicators of compromise (IOCs) or identifying malicious processes
- **rootkit detection**
  - analyze the kernel and detect suspicious hooks or tampered modules.
- **behavioral analysis**
  - study process creation, file access, and network activity captured during suspicious events.
- **cryptographic analysis**
  - recover encryption keys, passwords, or sensitive data
- **user activity reconstruction**
  - browsing history, chat logs, or executed commands

Rekall

- **open-source memory forensics framework designed to analyze volatile memory from a variety of operating systems, including Windows, Linux, and macOS.**
- **focus on modularity, scalability, and usability**
- **respected in digital forensics investigation ...**
  - as well as troubleshoot system crashes, or analyze system behavior in depth